

NEHAM JAIN

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EDUCATION

Carnegie Mellon University

May 2025

Master of Science in Robotics, Advisor: Prof. Ioannis Gkioulekas; GPA: 4.11/4.0

Relevant Coursework: Deep Learning Systems, Learning for 3D Vision, Robot Learning, Physics based Rendering

Indian Institute of Technology (IIT), Madras

June 2023

Bachelor of Technology in Electrical Engineering; CGPA: 9.24/10

Minor: Machine Learning

Relevant Coursework: Modern Computer Vision, Machine Learning, Computational Photography

SKILLS

Languages: Python, C++, CUDA, SQL

Systems: PyTorch, JAX, FSDP, NCCL, Slurm, ROS

PROFESSIONAL EXPERIENCE

Meshy AI | 3D Foundation Models [[Meshy-7](#)]

March 2026 - Present

Research Scientist

Sunnyvale, CA

- Core contributor to **Meshy-7**, Meshy's image-to-3D foundation model, working on **large-scale training, data**, and **evaluation**.
- Work on **scaling** 3D generative models and the **distributed training** infrastructure behind them.

Meta Reality Labs | Data Pipelines for Embodied AI [[Embody3D](#)]

June 2025 - March 2026

Research Engineer

Pittsburgh, PA

- Developed **multiview human motion-capture** pipelines; trained custom models for **pose estimation** and **hand tracking** to produce motion data for embodied-AI and humanoid research.
- Built QA tooling and **automated quality filters**, coordinated **1.6M video annotations**, and curated **2,000 hours** of motion data for **Embody3D** across body motion, hands, audio, and text.

Adobe Research | Scalable Pipeline for Training 3D Foundation Models

Summer 2024

Research Engineer Intern

San Francisco, CA

- Created a **pipeline** to convert **synthetic mesh assets** into **relightable Gaussian Splats** for training **3D Foundation Models**.
- Wrote **custom CUDA kernels** to replace **spherical harmonics** with **neural features** for **reliability** and **faster training**.
- Scaled the training **pipeline** to 32 nodes and **improved training time** from **6 minutes** per asset to **30 seconds**.

RELEVANT PROJECTS

SmokeSeer | 3D Reconstruction Through Smoke [[3DV 2026](#)]

Fall 2023 - May 2025

Guide: Dr. Ioannis Gkioulekas

Pittsburgh, PA

- Fused **RGB and thermal video** with **3D Gaussian splatting** to jointly remove smoke and reconstruct occluded scenes, decomposing the scene into **smoke and surface Gaussians**.
- Integrated **fluid particle hydrodynamics** into the **3D Gaussian Splatting** pipeline to accurately model the **temporal dependence of smoke**, resulting in more **realistic** and **artifact-free** reconstructions.
- Adapted **Mast3r-SfM** to perform **localization** of the **RGB and thermal** cameras in the same coordinate frame.

3D-Aware Vision-Language-Action Policies

Spring 2025

Course Project for Robot Learning

Pittsburgh, PA

- Explored **3D-aware VLA policies** for manipulation, studying how RGB features and **depth-derived geometry** complement language conditioning.
- Investigated explicit geometry for **sample efficiency** and **viewpoint generalization** under occlusion.

Needle - A Deep Learning Framework in Python [[Report](#)]

Fall 2024

Course Project for Deep Learning Systems

Pittsburgh, PA

- Created a general purpose **library** for **differentiable programming** from scratch with support for **C** and **CUDA** backends
- Implemented features such as **gradient accumulation** and **distributed training (NCCL)** for training larger models.
- Trained **Llama-3** architecture across **8 4090s** nodes on **OpenWebText** using custom **NCCL** and **Gloo** backend.